```
clear;
clc;
close all;
% Design Parameters
Fs = 8000; % Sampling frequency in Hz
Ap = 1; % Passband ripple in dB
As = 40; % Stopband attenuation in dB % 1. Low-Pass Filter Design
fp_lp = 1000; % LPF Passband frequency (Hz)
fs_lp = 1500; % LPF Stopband frequency (Hz)
wp_lp = fp_lp / (Fs/2);
ws_lp = fs_lp / (Fs/2);
[N_lp, Wn_lp] = buttord(wp_lp, ws_lp, Ap, As);
[b_lp, a_lp] = butter(N_lp, Wn_lp, 'low');
% 2. High-Pass Filter Design
fp_hp = 2500; % HPF Passband frequency (Hz)
fs_hp = 1800; % HPF Stopband frequency (Hz)
wp_hp = fp_hp / (Fs/2);
ws_hp = fs_hp / (Fs/2);
[N_hp, Wn_hp] = buttord(wp_hp, ws_hp, Ap, As);
[b_hp, a_hp] = butter(N_hp, Wn_hp, 'high');
% ========================================
% DISPLAY NUMERICAL OUTPUTS IN COMMAND WINDOW
% ========================================
fprintf('=== EXPERIMENT 1: BUTTERWORTH IIR FILTERS ===\n\n');
%[output:4777d2d6]
fprintf('--- LOW-PASS FILTER ---\n'); %[output:1dd3baf1]
fprintf('Calculated Filter Order (N_lp): %d\n', N_lp); %[output:2b993970]
fprintf('3-dB Cutoff Frequency (Wn_lp): %.4f Hz\n', Wn_lp * (Fs/2));
%[output:45e1877b]
fprintf('Numerator Coefficients (b_lp):\n'); %[output:1c062d6c]
disp(b_lp); %[output:2f8a758b]
fprintf('Denominator Coefficients (a_lp):\n'); disp(a_lp);
%[output:8929ec27]
fprintf('--- HIGH-PASS FILTER ---\n'); %[output:9526acc4]
fprintf('Calculated Filter Order (N_hp): %d\n', N_hp); %[output:980ee51b]
fprintf('3-dB Cutoff Frequency (Wn_hp): %.4f Hz\n', Wn_hp * (Fs/2));
%[output:9f41da7c]
fprintf('Numerator Coefficients (b_hp):\n'); %[output:897af61f]
disp(b_hp); %[output:5e9b4c42]
fprintf('Denominator Coefficients (a_hp):\n'); %[output:01e534a4]
disp(a_hp); %[output:1704a056]
% Frequency Response Calculation
[H_lp, f] = freqz(b_lp, a_lp, 1024, Fs);
[H_hp, ~] = freqz(b_hp, a_hp, 1024, Fs);
% Plotting
figure('Name', 'Butterworth IIR LPF and HPF'); %[output:6165f0d0]
subplot(2,2,1); %[output:6165f0d0]
plot(f, 20*log10(abs(H_lp)), 'b', 'LineWidth', 1.5); grid on;
%[output:6165f0d0]
title(['Butterworth LPF Magnitude (Order N = ', num2str(N_lp), ')']);
%[output:6165f0d0]
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)'); ylim([-80 5]);
%[output:6165f0d0]
subplot(2,2,2); %[output:6165f0d0]
plot(f, angle(H_lp)*180/pi, 'r', 'LineWidth', 1.5); grid on;
%[output:6165f0d0]
title('Butterworth LPF Phase Response'); %[output:6165f0d0]
xlabel('Frequency (Hz)'); ylabel('Phase (degrees)'); %[output:6165f0d0]
```

```
subplot(2,2,3); %[output:6165f0d0]
plot(f, 20*log10(abs(H_hp)), 'b', 'LineWidth', 1.5); grid on;
%[output:6165f0d0]
title(['Butterworth HPF Magnitude (Order N = ', num2str(N_hp), ')']);
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)'); ylim([-80 5]);
%[output:6165f0d0]
subplot(2,2,4); %[output:6165f0d0]
plot(f, angle(H_hp)*180/pi, 'r', 'LineWidth', 1.5); grid on;
%[output:6165f0d0]
title('Butterworth HPF Phase Response'); %[output:6165f0d0]
xlabel('Frequency (Hz)'); ylabel('Phase (degrees)'); %[output:6165f0d0]

%[appendix]{"version":"1.0"}
%---
%[metadata:view]
%   data: {"layout":"onright"}
%---
%[output:4777d2d6]
%   data: {"dataType":"text","outputData":{"text":"=== EXPERIMENT 1:
BUTTERWORTH IIR FILTERS ===\n\n","truncated":false}}
%---
%[output:1dd3baf1]
%   data: {"dataType":"text","outputData":{"text":"--- LOW-PASS FILTER --
-\n","truncated":false}}
%---
%[output:2b993970]
%   data: {"dataType":"text","outputData":{"text":"Calculated Filter
Order (N_lp): 12\n","truncated":false}}
%---
%[output:45e1877b]
%   data: {"dataType":"text","outputData":{"text":"3-dB Cutoff Frequency
(Wn_lp): 1087.8354 Hz\n","truncated":false}}
%---
%[output:1c062d6c]
%   data: {"dataType":"text","outputData":{"text":"Numerator Coefficients
(b_lp):\n","truncated":false}}
%---
%[output:2f8a758b]
%   data: {"dataType":"text","outputData":{"text":"    0.0000    0.0000
0.0002    0.0006    0.0013    0.0021    0.0024    0.0021    0.0013
0.0006    0.0002    0.0000    0.0000\n\n","truncated":false}}
%---
%[output:8929ec27]
%   data: {"dataType":"text","outputData":{"text":"Denominator
Coefficients (a_lp):\n    1.0000   -5.4622   14.6771  -25.1616   30.3497
-26.9537   17.9862   -9.0534    3.4018   -0.9283    0.1743   -0.0202
0.0011\n\n","truncated":false}}
%---
%[output:9526acc4]
%   data: {"dataType":"text","outputData":{"text":"--- HIGH-PASS FILTER -
--\n","truncated":false}}
%---
%[output:980ee51b]
%   data: {"dataType":"text","outputData":{"text":"Calculated Filter
Order (N_hp): 10\n","truncated":false}}
%---
%[output:9f41da7c]
```

```
%   data: {"dataType":"text","outputData":{"text":"3-dB Cutoff Frequency
(Wn_hp): 2379.7556 Hz\n","truncated":false}}
%---
%[output:897af61f]
%   data: {"dataType":"text","outputData":{"text":"Numerator Coefficients
(b_hp):\n","truncated":false}}
%---
%[output:5e9b4c42]
%   data: {"dataType":"text","outputData":{"text":"    0.0006   -0.0055
0.0248   -0.0662    0.1159   -0.1390    0.1159   -0.0662    0.0248   -
0.0055    0.0006\n\n","truncated":false}}
%---
%[output:01e534a4]
%   data: {"dataType":"text","outputData":{"text":"Denominator
Coefficients (a_hp):\n","truncated":false}}
%---
%[output:1704a056]
%   data: {"dataType":"text","outputData":{"text":"    1.0000    1.8915
2.8535    2.6014    1.8704    0.9534    0.3726    0.1030    0.0199
0.0023    0.0001\n\n","truncated":false}}
%---
%[output:6165f0d0]
%   data:
{"dataType":"image","outputData":{"dataUri":"data:image\/png;base64,iVBOR
w0KGgoAAAANSUhEUgAAAT0AAADtCAYAAADN0qIGAAAQAElEQVR4AezdC5hkR3UY\/hp5MctDI
DmQoFg2LRuMbMzLGGkJRLQ\/4LMcjB9gYkCAmlgGQYil2OGhYD6uA\/xBYGIp4WUgYRYEJFFk
HIOCiLHphTgIg6MYoxjZGDVmeRn88VS0mLX5z+\/ePTO1d7t7bvd093TP3v32dJ2qOnWq6tS5
p0497p1Tvt3+ayXQSqCVwEkkgVNS+6+VQCuBVgInkQRao3cSDXbb1VYCrQRSao3eNlrQZrcSa
CWwtyTQGr29NZ5tb1oJtBLYRgKt0dtGQG32fCTwute9Ln3f933fUJA3n1pbrq0E2uVtqwM7lc
AE5f\/qr\/4qPexhDysNnWKf\/OQn0zCQxyDe7373S3\/yJ38i2kIrgZlJoPX0ZibKltE4CTB
4l156abr66qtLQ3fxxRePJJfHGP7u7\/5ueulLXzqSrs1oJTCNBFqjN43U2jITS+Dv\/\/2\/
n972trclYdPCaJVpSt\/StRJoIoGFGj17NZYtw0Bekwa3NHtDAjw\/S933vve9KXB6IU18b\/
RSL1pYNgnM3ehRYIpMoXXesmUYyEPT7uOQxN6Hl7\/85elBD3pQesQjHpF+67d+q+zw9ddfn5
7whCckeWVC+9NKYA4SmKvRY\/DafZw5jNqKs6QXH\/7wh9P555+fvvGNb6T3v\/\/9pQG0nL3
HPe6R5KFZ8W62zV9SCczV6FFiezLCpv1Hq0xT+pZutSXw\/\/7f\/0uf+cxnEmO32j1pWz+lB
BZebK5Gb+G9aStcCQnc\/va3T9\/93d+dPvGJT6SPfvSj6Stf+Up6yEMeUrb9uuuu2\/T6yoT
2p5XAjCWwEKP3r\/7VvyrvZtmze\/SjH13iP\/RDP1SG0uSP65d8dDa9R9EFDboc8j1CS6bYX
8xpAh92mFIvk9NYmj3xiU9s3I9RbZ82Pfo8Ti541\/sQ\/Y0w7xN6IE2+UHwaiHq1My9v7+6D
H\/xgetnLXpae9rSnpa997Wvp4MGDCd1nP\/vZVBRFTr4trv\/aqvy2xGMIor10BI4fvsbYWE
dReWjo1qFDh8q7h+LSgyYPpcvHqw45b3cS8azTRFw\/o03S0L7jHe9IQvE6oI12GEf5wkjb7X
CcXObZ1rkbvV\/4hV9Iv\/\/7v58ouVn9C1\/4Qilr6Q40pFH0eQ3G17\/+9fSkJz1pZpdc7T
\/FAxBLs7JDS\/LjwXn6059e7pU1bZKDAw9UU3p0w+rBI3\/QHFDw4i688EJFSpDP0K2traVT
TqnUb21trTzMsJd3xRVXpDve8Y5pUf+0R7vH1cdIg3E00+bhSx6hU3\/7t387ktV\/\/I\/\/
sZRTENDtF7zgBWlUGfLXv6BftXAavWzSx0rrmlBOQfOud70rve997yvvZt3hDncolfme97xny
anT6ZQhBf9n\/+yflZvZMfBlxvCfbVNf\/\/rXl5dfw6AeOHAgUY4\/+IM\/2Cz7Pd\/zPYmy
ocnBpdhNohpy6qmnpjPPPLPcf2LsZDPYn\/70p6G7Ar\/+679e9tUJqAZ4eH\/6p3+67K94HY
b1+znPeU5JZllZIg1+htVj0vLwRnGz+Nvf\/vZ073vfO5111llllsnIeRJF8nIyFtpElRZc\/b
6BrvKxoz3b1vehFLyqv1mxHNy5fH\/U1dO6\/\/bf\/lujVjTfemG6++eb0h3\/4h+Vk9e1vf
zv93d\/9XTI2QaucvU\/0ytE9un3LLbeko0ePJviVV15ZVq8c+YqYSIwFfFmhLhd91gftnUQv
0TeB0uiZDbiTQoVCIbjNZnRp8tAEUHLpAA1aSsTLQPMTP\/ET6Zd+6Zdkpz\/7sz9L973vfRO
lLxOO\/QRPD8v\/\/b\/\/N4UxOZa944BBPe+880o+9o9KZMqf0047rTR6HkwKh00MyGMe8xj
RTQj5kUMA2UgPIrKIPHKIOBwN+coXSoODnE+kKwvIUVkPSMgbjQfi1ltvlVUCnngJy4SNny99
6UvlMp2Bksej3Ugu\/+MtzdaE\/Ho96ghjxYjQhXe+852JrEw42qK8ch5oCs1Q4ydd+2IFcM0
115Tt+A\/\/4T+Uy7af+7mfS7EVEqe98kya2vLQhz603BvEV934qR8NHKATvvCFLyyXouI\/\
/MM\/XE58Oqhdlp+5\/jEUMb5ra2vpL\/\/yL4\/zspQbBSEvdQL9Ux48\/OEP31x1aIdJgYz
+x\/\/4H+nFL35xWltbS7e97W1PYE3nyBO9crluB7FDQEbR2H3\/939\/Ykx42iHboDPu2gWM
XaQLxaUHkAtZyAP1vpG1518eQKtMlM\/1VX5TiIMtq8B4brbjPa5tUVbbThExG2hMVGDwzSoe
dEJ84sa+FYVCE0DJCSjiQg+bV4fgp59+umAkvOENbzhOiezrfOADHxhJP00GYVEAZUOIcMpDE
```

```
WNghPqIXv4oeNSjHlVmMaBoyYtiPeABDyjT48eeFFlEXCh+1VVXQUvj70ErIxs\/ZGtZt4Ge8
J+c5UdGzifSmobRbzyVEQJ96Pf7kjbBdsRm5BjysY997Bi2faAuhogXk1Ovra2l29zmNuX+Xf
4w\/s3f\/M3mchf9\/\/pf\/0uQ\/uiP\/igdOXKkxJXTf8Yhlsb01N7g2tpaSeOHEXnJS16S
1C8e8Ja3vKU0xOJra1v04re73e0EJTAUz3zmMzcnYXz27duXXv3qV28arJJwhz+8O\/LxjHnW
7nOf+5SeX8jC2NBNwADzgNxsYPDoH922gtI2TWH08CGT\/\/2\/\/3fZV\/F\/8A\/+gewSrL
7wLSMbP3SLEdxAk1AcHmAcg55RyfUWDVk\/61nPKr1gtuRxj3tcWa88YLyU0V7xpqC\/aP\/h
P\/yH5QpxO97btQ0voD+nxAwiwQAYiEhzedQDruEGhltN8EJxBi638nhwq9EYnFe84hWS0g\/
8wA+Up3Rm9zJh48fAqQftgx\/84I2UVLr4JbKDHwKmJICHEW2P08EdsE6UR78pGwNAXk4hKV7
ON5ad+gZ4NvLJUhgDykOUTw53utOdZJ0ADJI+oEOPIPjAA8iW7MUZdDyliW8HFGFtba1cTj33
uc8tye985zuXoZ+vfvWrgtJY\/c7v\/E6KehieH\/3RHy0PHqKP2vjHf\/zHieeo0P79+xN90
QfyY0B4c+Kx1fGd3\/mdydLsZ37mZxQp4XOf+1y5xCsjx37Ingd5LJpMnJbz2oEvz1G\/yQzN
ve51r1Lv\/vk\/\/+ei6Tu+4zuSh14f0atXhv4YE6fK4h5kk6C+iN\/lLncp++7BPXjwoKSxQ
O7GK0Bfv\/d7v7f04ixHtZl+CtXFe\/v5n\/\/5xFCvrR1vjEdVVBRF6amee+65Zb\/UwYs0l
uDZz352WdQzzBiWkY0f9aHVthiz0CfbO9IDyGWjSDK56zsckEvQCA9tHOaog8FUd54PVx9Qdh
goQ2fJJAAv4x37weLo8FMngOMLgq80eQHRtsgXnhIdRkwgZl8KINNyIvKf8YxnJDORdOEjH\/
nIcu+ovlfG0KDZDnSS4UN3t7vdTTAX8AAwztocFUgjqBCMkJGO9gRdPbRkoKBm0r\/4i78o+2
\/5PKqcmdMgxkyJH+WhRPmAKm9fU34dcqU1HvX8SeLR71B2YUxMvAV9CX73v\/\/9A008H5Fv
fetbyYP6i7\/4i6LlBrq+WBmUCcd+9NFSjmHhpSljvOnW2trapkE8Rp4YxrW1tfLhspelLZ\/
61KdKo2epZ7ZXntIPBoPSeDCgvHcegPJ48QS1xca+uj\/+8Y+XOstDk7+2tla2GQ5GTTTGJiZ
J7fmu7\/qupE10yMNHn5QfBbwO4x6g7eQxjN5zV9c9Rla9xoduAhNplLfC0g58TfKRXg+Vz8v
Jz\/WJ\/KTVIdpf5+3ZpkPqjr5pS5QPW5Hnw+VHHrwJ6BvbQuboozx+UTc88rZrGzqg\/adce
+215drfA2WwnRBZ7srECGFT4PXEbJmXuemmm5I9FI2NZQtFFAfR+LzMtLjZiZIEDLP00\/Lm0
XkAPXxOzfAZpjgUQb\/kaQflQzsN4DFNuXoZBivSQoHEf\/AHf7D0FBiKL37xi5JK+Ou\/\/u
sy9KO\/wlmBtvzMz\/xM+vM\/\/\/OSJWPFIFo18L54j\/YgLfNs0oszyCYbHqBC9vkYB3hT4
BHmtPmSNk8\/7bTTSq8+T1tbW0tPfepTy+WnsZ9GJgwxHeL50gtQN0p5ncNwuuV5YRToejxv4
rYkhMrJ47nBcxinTzFJ0w9twyMvy5vzPHEY2Ad52lLf15M+CeCFpzpNAspaRVl1wpvAJG07BX
PWn7B4MSo3oNIwCiG99rWv3dzPoJxmO0YyZsRhDbOsRWP5ajmoU4997GNLUksTcWC5IZHHI1x
mMDlE+wxWfWII2eR5lCjK6CPDyauOpRLPyGQTNNOGvB\/yHlaeoZFu2WlSgwPyZ\/A8kHk6g8
Tb+OY3v5mMHVrGyQNriase42pVkOsJOn2M5RwvTBn1xENkuYwmlrfaZGln6Xf22Wenyy67LHm
YtUsbLBfpmTrRqoNRxEPdvElp+qicZeza2lrCy3iEfl166aXpPhv7ZmgZwFGenvxh4Fmw4hmW
l6dpr\/4GeKbIA\/ze7\/1e6X3m9E1xRomRoVv2gPV\/WFlyogvD8kal8ZidtitL1uhyvRUPI
HPGT\/8YKbrMKyMfNNLk5TDMAKOtg0lAeTxjrxBNU96j2oYHYNtOgWBIgGZTcRAPN2MINMIyh
QcjFLfEDfdTmVFg0BkHbvMomibp3G315+A0xoA1KT8LGspEMfAa5tnK5ykQLrlpK0VFHxCylS
6fbBzkRP5Ow1zexhY\/BkWbfvInf3Jzo9myO07Y0bzmNa9Jl19+OTTxshgGe202xSUyKsb+p3
7qpxLDldcjH+iTmT+8MAZJGf00fmgYWHWH1y8NeJidup511lnlCa40bbjrXe9aXnsxKUtTXt1
4MqgM49raWrLfR5b6ic4KQzz65KE2wcjLQbvs4Vka5+nDcA+kdtbz1BnjrV0BPLM67TRxxpvj
oWxel7EwMUnXft4w\/aOH0pqCyYI+e65jvIxRXp7RjX5FaLw9DwxlyEZa5Avpg\/bnvMbhVkZ
krJ\/Rhu14b9e2vL5TosES4wFRIWWRxhjac1CpeICGscoRHxYqGwMi316QTezACQSEUg5TSLT
LBB7IePhMEvqYt89MY\/kTaWTJMxLypBhonkB4PejIlhcC3wnkbcPHTG2G\/Sf\/5J+IjgT1h
ze0trZWbncEsQMNd9TEbVEIGZ21ta0Nd\/XQF32U78GhV2tra+U9PWkB6uLlRzxC6bwH8iPTH
\/mRHymzLAc7nU55gsdDlviP\/tE\/EpTAe3re85533Emtup\/\/\/OeXe38l0bEfqwy8j0XL
A7bAGQsTQ8RHhdqXj+8oulmn86TIdRxfumX\/lfFixMbR1vPIhYdNdvKEnnuG3GrQUpMu5XqL
Drzyla8svVey8ZHY0AN5+Li10MQ5Qg\/wCRkzoJwlaeN4b9c2fIG2neKkLRrkYeSShvIhCmDg
5AWoJPKUx4eQCC\/SPYQ2k6OMB4uSmqUJUTpcGmGOmhXrdSsXEG0NGn2I+ushwaEHqkiUnAAA
EABJREFU8Hr+sDg69ACuf\/qp\/pCBOsW1AY+IS1OOsRACPPSTd0GhnDYqZ5mobADeygsjLfi
ilyZEI108b5v0KPuqV72qPBmXNgzwQRt52hn4DTfckC644ILyEjRPSrrDhBg\/cWX1Ky\/35C
c\/uTSelF4f0QF1aS+cPmj32tpa4v2arXlcJj+nk2joVXgtykp761vfWrYHrk71M2h4BZxzzj
mbNNrKqFoSmmRN7sp6WIUmcF6F01+ftsJTf9Snr8ZGO+g5\/tF+aVYa6PEZBnjghQbA8RgG+O
PpkrJ69Svo4MP465tnjE4yimDYZBzlhcEz+qGN0qr6\/7iUm3YcOHCg\/PArXB6aKJO3RZo8o
H\/6Gfl5Wfk5BC16eJ6HZ\/CAy0ODNtLrvNFFXoTSlM2hXN7mCfPCzUKWFyy4xkc9cGm8CTSR
vpdDD7j+Mfg8JA+iGU1a5MFXGYwrAxJeQr0vDL+rP\/a4zL7yvS4oZPCETYGxdIocxtW+W74f
xFPgZZO3yTXXNcsuBsMyGcClNa17megYZ7Jk4JepXcvWloUZvWXr+G62x+zDu6i3wQMpr56+q
nEPn\/2lOLCJfpjcGCHXdCxfI52nKk0emkjfLuS9MFbKo7W\/ZAnuRFicMYiDOd6L5R8DKc+y
0euFViW8GR6iNHmrBAy1w0UTjQlnldq+6LYuzOgZCAOSz7I6S7mlyUMjbRZg9udBAfgseM6Sh
wc1XPAI95LBIyvjaTkSyydpiwZLZV6gJW1et31IcaG9wjCYkSZcJWCwLffo1Sq1e1FtDV2kjz
s0epM12YBYypptGSMAlyZvMm6jqc16Zn7LFQCXNrpEm7NICVBAk5xrOvnhwawmQJ5leG\/Rrz
B6jBsjF+nCyKvj4i3sPQks1OgRH28mPJsIpcmbFViehNKbAYctWQ4fPly+xmOfp4Xrx8qCrGY
1NsHHJGcpG69OxcEOYygv6CYNefWWeU4iGbhJyw+j1\/9WR8brSC4f8homx2VJW7jRW0THt1u
```

yGBTvJjopbOGJaTsZkBWZzXrsTHYx8UW4U4PnLtuwKxJ0QvtjuQsPiDzxHBfXb\/3fTkZt\/p
YekRe5kd8ywlyNnuUKZRA27TxaZZrSj6Ibt2QxIB\/60IeS904tfecI5ZF\/E\/6Wev\/23\/
7b9OY3v7lxmSZ8d0pzySWXJLIis1GynjSd4THGTlUnLTuKnofndgAvj3cfdLy9+nJWXuhHhNI
C8jT91v9l0hVjepy+vPSl6W3gbW\/bdd2Zh77EuMwqnKvRs3fjbpTPzdi\/G6fklBaNtzwsTW
bVwVF8Pve5t6Wf+7nHpgc\/+MCuwhOecCCBpz61m172svPTddcdSB\/\/+IF0\/\/vffynggQ
984CgRTp0ebw5MzaBWkF7x8NxHpHO17HT++ecnBtGEev3115cfgnVVCJ2TXvfi7PkCuDR5OTj
xPXDgQPmxzuPCu90tHXjCE9KByy47MW8c\/W\/+ZjP6YTyOpXnTwXvJB173unTg8Y9v3oZj5Y
\/rx4zSLrnzndP7d\/HDuvmYjcLnavRUSgmdmFi+uDrAsA0DtGicQOUztfRpIF+m5Pg0vOZZZ
jBIaTBIaTBIGx7V\/vSSl+xLT31qSve85770jnd8OXnBfjfBPbpZ959O2LvjrfD6dsqfXrmU+
9M\/\/dPl62uhX4wh3pbRrqx4yO0dOjjTBnl0jdepLIBLk9cI+v2UBoOU+v2U1te3L9LvpzQY
pLS+ntJgsD19E4p+v6Lq91P6sR+r8N34\/bVfS5T3TF9yfs97dqMFjeqcu9HLW+HqAsM2DC6+
+OKcdEd4vjwJRvW0U0+9Jl100eFUFCkVRUpFkVJRpFQUKRVFSkWRUlGkVBQpFUVKRZFSUaRUF
CkVRUpFkVJRpFQUKRVFSkWRUlGkVBQpFUVKRZFSUaRUFCkVRUpFkVJRpFQUKRVFSkWRUlGkVB
Qp9XopXXDB0XTuudXHMrX58OF9G3ttZ6Sbbjojn XHG7sHZZ5+tOTMFHhfPi9fF4wojFaG3HNA
0rXSUXjF2wSOnydPl073QSbi0qeDQocmK9fuT0Teh7vd3x\/ANBimtrzdp4a7TLNToLaq3lie
WKZYrAC4tr\/+Od7wm\/cIvHE4vfGHadXjTm1J64xuPbizBPpf+9E+PlIYw2voTP7F\/42R1f
\/nNOV8sWTTc\/e53j6bMLORlhfcfxiYP5aGZWYWLYrS+PllNn\/rUZPRNqfv9xRs+Xt5g0LS
Fu0q3J42e5YlliuUKgEvbVUk3rLzTqYzw+963VcBydyvWYkstgfX15s0bDJrTNqEsipS63Yqy
30\/prOoPMlUJ7W9IYE8aPZ2zTAnvAS5tlaDbTYkHqM2DQdrVrRptmCVYulrCxnJ2VOj93FnW
uxBekyxxB4PtmzQphdmy261KDQaV4ev3q\/gifweDRdY2UV171uhNJIUlJe71Uup2q8b1+yn1
+xW+6r+Wrg4yvBxvXy8mJ7g07yV7N9brYytn+NbXmw9Pv9+cdhJKhq8oqhKDQSpnzPX1Kt7+p
tboLbkS0N9Op2rkbh7MVS2YzS9Pz0FGfoqKM2MoTZ5rLd7Y8PEBeSsF6+vNmzsYNKedhNJmdS
wVlLNHAuAnOSzc6MV9PEua9773veWfnbPnNunVBQcU7inhA+oeQV4PfJXHua67q9yXk6Lt45a
49cOLfn9+Iun1UjJrRg3r69VydzCIlD0QTt6FhRo9hmkW31DjKTzrWc9K8RFIyyIeQRg3BtGt
dR8bAHBpk4tnOUp0uyl1u1Vb1tdT6vcrfFV\/eXSWt76uYyyjH3Bp8tC4X+lScOQvfdjtVk1c
X6\/CJr91I9ikzCQ03W5KN9+cUrdblRoMUrnc7fer+En4uzCjR6EZJkuWnX5DzQPhWkPctxJ3
+dRDYgybfHAA3SrBXvP2HC5ZyrowzFMHcGnyTGDesrjssstWZ5i63a229vtb+DhsMBiXO5u8T
qfy+Iqi4jcYpNLwnaTL3YUZvUrai\/ll\/Lxv6b3LqFFa4KsYdjrHn+a6FrWK\/cjbbNKKQ4w
IpaFh+Gb1dg5+C4ELL9yq5uD2fxS8JF5fL4OF\/NjnG7bc7fcXUv2yVLIwo8cbs2zx6tGsv6F
m6epFc+9YhmDzNzC28MitQn+pa5nAH1G69dZb06g2Pf7xR9JDH3q0bHxRpPTxjx8ZSTuKx6Tp
R49W9ZWVtj\/bS6DbrWjW16uwye9g0IRqNjTd7vDl7l6YRRtKaGFGT3vM3pa3s\/yGmmWz\/T
0fKggvQV1NwF+O2s33WvO6P\/\/5z6cvfvGLySfO8\/Q6\/sxnfnGza71emvu7ufN493azA3s
Ryb29waBZD\/v9ZnSzoup0UuLxFcUWx6KoDjn6\/a20PYot1OiRIcMUS5kIGUN5o8ABiD0fkJ
\/0Mni+4GI\/z7uVefl8OZvjOY2\/zbqb77XmdfuLX\/6+693udrex79r+7M+eXr6nqx8f+tD
+ub+bO493b7Ud2LczpmAnJ\/l4LQ30eltN6fe38HHYuNPeceV2mme5O+yQY4\/v9S3c6E0zTg
xaGEgnsfbqwuBZMsvP+Q5bzg5L27dv366901p\/h\/Z2t7tduu1tb9uoPVddtS91Oqn8593cO
q8dxE+ofx7v3mq4iWwWJ\/l4LR10u1WTtjNm3W5Ft75ehbvx2+mc6PWtr6fyFbZJlryDwW60f
qo652r0GKYmrxuZ6dGhb9ILd\/ouvfTSxMMb5iX6uICPDNjrA3BpTXivCo1JOtq6ahOzcZ7VS
X7IYKnCTqdqzvp6SoNBhQ\/77Xa3UgeDLXw3MApV9\/qKojJ+\/f72Ler3t6dZEoq5Gj2HF4c
2Zrvw0rxe5DUj9+oiDS7NVQX0TeRy88bg+Huq\/lYsgxkQS18fF4D72ACAS2vCe1Voer2Uut2
qtevrKfX7Fd7+LoEEHvawrUb0+1t4Hcu\/YNPv13MXH+90Kq\/Pfl\/UPhik8nrLJF5flF3Sc
K5GL++z2d3rRXXjxtBJcykVTV5mFM6Auc4QhjPCWPoqxwOMdLi0vQarenfPmNuWmMdJ\/rzHu
BH\/Xi+lTqci3Zj0K2TIb6+3lTiObotqMVi3m9K3v51SUWzVVxRp5JJ3MNiiWwFsYUZvBWSxc
k3sdFIqiqrZg0FKqzQZm4hmfZJfSWLy3\/xABT45hyElut0qcX29Ckf99npVzvp6FS7T7wtfe
OL1lqKojF+\/v0wtnagtCzN6MbvXPTrenTQzP5qJWt8SJ3rZ7VaCKIqU+v0KX4XfaU7yZ90ve
75WCF5XBHBpO67nYdkSdzAYza4p3WgO883pdLaWvJ1OVddgkMolry9gDAZV2gr9LszokYnZ3V
L2wIEDm3\/LAC5NHpppwHWH+kGIGTv2+uDT8F2VMgxftHWVvL1o826Gc3tlsdfb6la\/v4XXs
V5vK2WZB6\/brby+othqb79feX0Hs7dPtnKXFluo0SOFYbO7NHnTQHiKeVkztRnbzA3g0nKav
YR3uykVRdWjfj8t5TLXOJmYYiIaF6JDX\/Vovr\/ucM7tlcVOp2r8dvt13W5Ft75ehcv8a4bd
OEhMRbHVyqLYwlcAW7jRm7VMnOCeeuqpx7Gd2+x9XC3LFaGL3W7VpqJIqd+v8GX5tXVxaOPhj
8OlWZ3kz6J\/+R3OHN8x7263YrG+XoWjfi+8cCtnMNjClxXrdFKicIxft7usrRzZroUZPTO3G
XzUDC8PzciWDsngvd1www3poosuOi53rrP3cTUtV4QeRouWeaVknGd1kh\/9nWfoTxqOemc5f
ze5TnP0IQ\/ZbNbR9763fE\/66Le+tZm2Sf\/4x2+mHf3VXy3pNvOOjH6\/Ot7VTsf+4d2k3M
xo7na3dOTd705H3\/CGYy1YjWBhRq8+05cz\/ic\/mdzTu\/e9751e9apXJTSTiO3gwYPJa2j
e0KiXy2fsHM\/pZjb4YxRzkjpCiScpk9MeOHAkPf\/51QcC+v2UfvVXjzZ+gHI+OZ4\/1Lns9
hJukoz+5Li0A5ddlt7\/6U+ne3\/kI0Pfc\/7yV76CrATvTufvSn\/64Q8v0\/18\/V3vKst\
/\/RvfEC0hpz1y7rll2r63vrWky\/OG4fm72mXBjR+8h9HOO00\/j5555kYLVuP\/wozeKHEw
dJdcckl66UtfmrxpMYqunu7wQtpO9gNX8YMD2ynwL\/zCp9O55x4hmvSSl+xL73jHlxs9RKP4

zuODA8bcaf2LXvSixOsrG7vxA5cmD81G0tz\/D5sQj0sbDMo2nH7llUPfiT79tNPKfD\/enc7
fp4anTkdWOvVLXyrLn3rHO5ZxP\/ID9mWrlTNuuqmkjbxhYf6uNl4A72G0i0jzSqc2rALsutE
LIX3mM59J\/i5CxPPQe5qxLPZ2hS+RXH311enCfC8kL7CB5zN2jm9kbf5fxQ8ONFHgF79432Y
fX\/Oau277AI3jOa8PDjitd2rv9D7GFi5N3mYH5ox4PdFrirZKAFzaZrXH9gz2HT6c9l9\/\/
QnvJucP+7B3nlO3W7Liwcnfd5vblHE\/4gG50dv\/n2K\/acIAABAASURBVP\/zCfUEXYT5u9
p4Abwjf9Gh+lcFlsLoXXfddclnwW9\/+9sPlZsPCsRy2Eksj8Cy2CtmHpinPe1p6dMbSxCfl6
K497jHPU7gMyyNwi5aOUbVlyvxFs3+bZV\/GO0jHrEvFUUlgv\/5P\/elyy+fjg\/e8\/rggN
bx0mNcI5Qmb1Hg7R4TKV0CcGlR\/\/Vnnx1oShvbKVuRhtjDHrZFOBhs4cOwXq9KXV9PaTCo8
PZ35hJYmNFjqBxWMFJ18AFQnwUftjc3rMeUMn8N7fWvf33y\/i4+8szUZmwGEMClDeO1V9M4K
N1u1buiSKnfr\/D290QJ8CzD6MLrFNfv318lra9X4SS\/vd4Wdb+\/hQ\/D8pXLMp9EDWv7Cq
UtzOjZozmUXVkIJRMyYIzVrOSGlxnbzA3g0mbFf1X4MHzR1t3+EotJzzgIo03bhWiV2Y5u3vn
X5Fei1tcnr67Tqcps6H+FjPjtdlPqdqvM9fUqXJXfTmdVWrq4v3tLgXl6cQCRS0iaPDR5elPc
kohBZVijjBmbQQXwSD+Zwm43paKoejwYpF29tGxsrrjiivK0nadvzKuWnfjrDRo0tiusAE6k2
ExZCHLNxuHD4X3H9km3M1zDWtTtVqnr61U47jf39tbXx1G2eVNKYGGe3pTta4vtUAK8vW63Yl
IUKfX7Fb4bvwyfyclEZB+XYRsG2oZm1isAfKeFw7FPvL6e0mAwGZtJ9\/W63Yr\/brvnVSv23
O\/cjV7M2k7mHDY4dKgrujQfBPVQTCLh4I2fZVB+5SXPg0\/Cd6\/RLuMnqPLDKQYuh6X0zHu
9LbXo97fwSbF+f\/sSubfX7u1tL68JKeZu9CgwhXba6rDBoYN4HTwEk7Td8sjfRfVuLV7enyy
KomTh8MIprzwAl1ZmnoQ\/nc7xfz7SxzFOQjHsrMu8tU6n4nEwe8E+\/2PdnU6VX\/\/t9bZS
BoMtfBTW66XU7Va5RVGFU\/62xU6UwNyNXlTJizu0sR9i\/y3SdhJaHj3jGc9IcUDBaAI8T8Z
3b\/V7HPR6KXW7FUW\/n1K\/X+Ht7wQSsFeAvN9PaTCANYdOpzktytzba5e5JDIzWJjRm1mLN
xg58PA3FobdvdvITi4j8\/zyKzDS5J3MkC9zeXuDwcksjSn63uttFZp02dntbpVtgvV6KXW7F
eX6ekqDQYWvyu9gsLQtnavRY5ycylqKBm7\/bRigQ9NUUne6053KvxF7v\/vdr\/w2X31PLze
IOd6U\/16k63RSyv\/8QetATDHK3W5VaH29Cpv+Wh43pQ268CzFV22wul2tXko4zujNuoX5kj
Zw+2\/DwNIXTdM2fO1rX0vXXnttspT96Ec\/WhYriqIMm\/7kL9YvA77TDw406YOPEjz0oc0\
/SnAyfHCgqb6UdBdeWAblT79fBnP76XZT6vUq9v1+Sv1+ha\/C7zRGfkH9mqvRm1Uf6u\/exj
u6\/saCJSyAW\/KGt5gvZ3M8b9Ne\/ODAqA8H5OlvfvOnG3+UYB4fHIgxcKoeXr\/VgHjdYw\
/apQl7va2mHMwONLZSh2O93vD07VLrexLb0bf520pgoUbPCWosR0PZIxy3vHVAEd6hk1gfCvC
ubr130ry\/O2w5OywNn3Ev2y8yL\/9qxiLqzT9K8JznjP4owbw+OGAiW9k\/9t3rVaq3vl6FT
X87naaUx9PlexI2Y4\/PbWMTSmBhRs8dOp+Pcss+DFgeHto42W26vOXZnXfeecmfEMQXwKXJ8
56t920ZWQCXVpfNxB8c2D\/9i\/te3h8Hs\/zgwLh6Ii\/\/KMHhw\/vSM54xvG\/z+OAAb5x
Xzju\/wx3usDksxk6aPDSbGcuG5Eu39fXmret2m9PmlN1uSt1uldLvp9TvV3j7O5UEFmb0LEl
9Pur888+fqqH1Qu7\/MXL3ve99E3BaKw2dayyWSd67BXBp8lrYkoB98m63iq+vp9TvV3j7u40
Eer2UOp2KaDCownn\/8vY6naoW3l6\/X+Ht78QSWJjRs+y0\/Jy4hWMKMHLhLVoC56R5HjzPa
\/EtCcSWUaeTyr\/qNxhs5c0L49E\/4QlPKD31W265ZbMa3t2iPyK6WfmkSLc7aYmUcg9x8tJ
bN8w3yu5\/xjM2ftv\/00hgYUYvli6UmnJP09i2zOwl0OmkxIkYDCreO7sZUfFo8msispR9+M
MfXn4L0auIXlVkDOU14bGrNBdeOHn1nc7kZfIS3W5KRVGlDAbpjI2Jo4q0v5NIYGFGj6Fj8Lx
\/S7njACPCcQcZk3SopZ1cAt1uSt1uVa7fTwv7Gou3c8JTj3AlDB5RdbspdTqw5tDtNqcdRZn
tSez\/0IfS\/ssvH0XZpo+QwMKMniWNw4pQ7nooD82Idg5NdgIYRhOeE7n+EHnwPK\/FT5QAb
6\/brdKLIqV+v8Ln9WsSNNG5qhK48ZImPmm9xl954IaAA6ycR56vzjyPfigH4HneWJwBGkswJ
LPTGZI4YVJ9sCYsfrKTL8zozVrQlPOzn\/1scjHZxwyc+ElTD4V3tcXHBgBcmrwWRksgf4bnv
cx9+ctfnnxZh7fnbxdrlXG0vJUn3hSMe+iCydQ72c961rM2\/+gQI0c\/8PfBCyuOMKz0gn7Q
EwCX1qjuXq8RWUaUUqdzXHTayJHXvnbaoid9uYUZPUpmFjebjgMzcpNRceHYia29Qh6iB0ias
t7SOPPMM9NZZ51VfpDgnHPOKd\/ckNfCaAl0uykVRZW\/sWVUHmxUsdn+0gVGyEm+60bu6xk\
/4+g+pTw0TWu1JGas6IIyrid9+9vfLl9TFPdxiuBva8WBmslS3o51pdPBpjl0Os1px1F2Oulz
b3vbOIrF5m20J6\/wcHx0NU9cEnxhRo9Cm8V9XsqMa0YGcGnPec5zSq\/NjN3E8Hk40HpoPCA
eFA8RuTJ+YRDFgTRhC+MlwNvrdiuafj+lQ4cqfF6\/cZXJeM6jDvpBT+r8Qx+EO9KV\/Pi7SQ
fiAKQomlCPpTly4EA68u53b9Hc\/e5b+KIxciiKstZlNngauDCjxzC9\/e1vT\/7EHwOocgCX
Js8D4ESPAZM3DszuaN3Rc+HZHwu3VIoyuZLneOQLP7SxEczoLgN85CMfSb4UvAxteelLryeeE
p73vI0H68i5JT6rn7i+xODwuLwOyDvD\/7rrriuXvfRCfBrwR+DD04\/yoQO8QUYu0oWRV8fF
A0bqyv796fr\/9J\/S9S97WWo0dkH\/4z\/ejP7664fSbepL8Lv44uQvtzVqwwieOy670adrN
rYVnnjGGWlawxfynme4MKM3607wBr2F4aGxRPG2h72dJvV4IM4999x05ZVXJheXlwGe8pSnpF
\/+5V9ODPkytOeMM55QinLfvsPpIQ85M5FZmTCDH4bH377wEVhXVUxaLo8bU15ZURRT10IH\/
FU8\/NUzNaNjBfV7W1257LL0RPDEJzbTp0loR\/A8Tl\/we897mtU9gt+sdO7Z116bvvshD5m
pvhwbipkFCzN6Zm7L23wTWS94gNLkoTH723ORF+BhiH1AgzPY2HDiDTIQFBvAeYv4KYePEOS4
OEV+xStekewDtfC2oXL4rd+6JL34xYfTBz5wOP3Gb1wycyVm5Hi2tjjiYrnQeBhP4zQMHDI4n
Q19cEgRdAweQ3rVVVeVe7mRLgwdiOWutIDIE89x8VZXhuuHcRoFni1yI79lhIUZPZ23JLWUtZ
kcSguXJi+U1iyNPsDD4OEABG15FHnDwny5Evn1NIOi7hYOpFEyeP7zzyzzyCotyb\/cWNKH2N
KgOyY9Xh6aaC4DWl\/Oygt9iFBaQD1N\/0fJqE0\/UX\/IK2S5jOFCjR4BUFLKmoM0eQyf2T9

XWul14BE+6EEPSpa3Zm4AlybP\/pCPDPAKAFxanU8b3z0JGJfcY4tJUOiUPzz2Ji3k7fHw7Os
a\/3oZB1wMIp72rbwDbi8YHb2gH9oD4NLktbA3JbBwozcrMca+D+UFZnMeIf6MpmWwjw0AuDR
5+VLZwyJtnsAgq79eF8\/EAw7geRs8nB58eULxPB+9PADP87bDPdi5sSGPvAx++AJ4nqcd2iN
PKJ7no5cH4HlejpOJPVj36ezJ8pbcn2OQnOTz\/IcZr5xHjjv8+PrXv56MtboDQuYmVROieuw
h5vzphfFRFsCl4U82dV7SZw3LMCZN+kSe9XE3ziEjeM6HfqCXLxTP89HLA\/A8b574Qo1efXB
1NmCYUMZ13LLFUjc8xjB4UYbXGHlw6QbNXqCHy0NmL7E+EOhmBR5uD5n6cp7koO0uwwK4tKBx
OddDqv1C8chDh145AJcW+eNCfXVp95WvfGXCW7vIIxQOH\/zwBXBpwVM7tEdZoXjkoUOvHIBL
i\/w8dErP27KMNI4mLHtpDB2DxGsnu7zMONzYa1MdGLsol9Pk6fLpR5SFS1uUrizLmOjzONBO
z0tOY3yNs\/EGcGlBQz\/oCdkKxSMPHXrlAFxa5M8zXJjRo8SznN2nEQqPgPA9XGZ9ByY8jWl
4bVfGAMYyifeS0ztttu8x7PI05WKILMmUufDCC9NNN920+XbBuLLoj4NaRL+97hcPvTh5MDhI
x\/GeZ7sYv2iDdjCIDCN8t2BRumIMlnFM6nL31sypp556XPJu6ctxjZgisjCjR4kpMwWfxew+
aV8ZXdch1J+XzR+2PH2nuD7+9m\/\/drriiitOYKVO3g2ayJQG\/8IXvpC8TeBLyuLgq1\/96
ubbBehGlUW7ExjHe5bt8qAztgyL9hoThp5h1QaT0XaHVcrNCxatK+P6QR6jxnuWYzKuDSbwG2
64IV100UXHkS1D245rUMPIwoxevT0UndAinUFkGCM+r1C9eDM4lAk+D+DFgVG8ox3yc1z8zne
+cwqjJxSXHpDT53jkNw0ps9PO8CqVy\/nluDzt0B64UBwekNPneOTnoT1Zk5ClNc+TEeR9O5B
wem98cvrdwKMP2jJPXcn7tptjkrcjxw8ePJge97jHJXLI0+EhozouTj\/oCVwoDg8YVzZo5hE
uzOgt++w+D+EuM09elf09F4MZnUW31QNkHyf20GLPrcnp\/ezaulycdntMhknD3qb03dAR9c4
DFmb0NH4ZZvfwLmMJo127AdEOdee4eL6ctYQRlx6Q0+d45G8XerjM3Lwrxianz\/nlOBrt0B6
4UBwekNPneOSvWhh9WISuLOOY6PfVV1+d7CuPGruQkfwcF6cf9AQuFIcH5PQ5HvnzChdq9HZz
dlf3sCVK7mLPS8h1vsPqjLRhywDLAun4BB08YFha5NXDeLi8AVM3eMP4RJr6tSPnJy5dWtDBA
4alRd4yh4vWlWUdk5tvvrl879dVHrcs3ETwEWCrA8vwYeMbafSCfuTjLC5dWtDBA4alRd4sw4
UavVk2fBpe9q5GXVKdht+0ZZzqugRLcQBcGn62Ae51r3sl+yjiQnHp4ujQKwfg0uRtB2buSy+
9tHyhP5aVeRl88MMXwKWhUb92aI+4UFy6ODr0ygG4NHnDAE1+X9BDFTDp9aVh\/HeatihdWaY
xqcvsPve5T\/kRDFdOgGvgJOdyAAAQAElEQVRebiLYB3aX0fgaZ2MJ4NLwoRf0g56IC8Wli6N
DrxyAS5M3bzipjJ59CUs6G+ZmLXfCYhDmLeicP4VxCdYMCuDSgsZntpxmMgJC8chDh145AJcW
+eNCM\/eNN96YXD\/AOwAPDx8+cHwBXFrw1A7tUU4oHnno0CsH4NIiPw\/V5foSj8HDVIdDhw
6l3RiXvI2L0pVlGZO8701x42ucjTeAS4vy9IOe7FRfgt+swrkaPcpNEDq9HSxqdreki4eMYs9
KkKP4eHg9xPW6eFrRDnhePsrIV1Y8z0cvD8DzvHE4hXRQoFwODhQs6ZTFL\/Lg0gK0Q3vkC8U
jT4heHoBLGwZO6Z3W86aG5S9L2iJ0ZVnGpInM6XB93I2z8QbwnA\/9QC9PKJ7no5cH4HnePPG
5Gj0Pkq+fRAe4xzo4DIYJJcq14d6SgDt47uLtrV61vVkVCczS6A3ts9khjJzLqOHxuZ81tECb
uOclEJOh15ps4u\/5DrcdXCoJzN3o5b2N5YJXv\/xdhDCAXuzO6Vp870mAcbOFEWNuT9VJoP3
VSIsQHfq9J4W2R8sggYUaveiwmd4+Eg+QAXQzv1X0kM7eDO3n2MIw5tsBOvR7UxJtr3ZbArti
9PIDjvgs1Mmg6Ls92MtQP6++9eiWYSRO3jYszOjVDZ0DjpjxLXt3OgRel4mHqR62XuTx0rV0f
PSjH53cj4KTD\/kdT5VSjFmT\/Ve88MSvzifiDB6vnndv7F2Q9mbIuDJRdpahvtZ1JOJksej2
zLJvs+ZFFsbV+MLJh\/zq9cxaV+r8Zxmfu9EjIArl4qEXySk7cMAxy47g5eKkb8Thn0PrRZLO
Fviu2aMe9agT\/o7EFsXkmKsXeOI9rLSHgsEz2dneQPOkJz0pOcVlBMUXCa2uNJO28TSuxrdZ
ie2p8MIT7+2pZ08xV6NH0X0QUrOHfdmWMQwwg5hJ0LYwPwmYsX2f7zGPeczMK8ETb3XUmcfdv
Hq6+OZ7lyItLI0EjKPxNK6zbhSeeKtj1ry34zdXo2dGjwOL3PMahi\/CG7NMe\/rTn17+qTzG
VpyAwhuVBiJdHsjzGee3vvWtJQ9G3aCdd9555VIRLainocsvaeMRBl4ojqdQ\/SBvQ72817fU
Aep1Bz9t1pY61F8HquePigdfbcsh2qKcwwczuDrEVxnIv9WVgyl\/dazpeC67rszV6DUV0iLp
vDdoicXwugVOud0Xi2Wx0Pu50rWL8fiVX\/mV5JPWynh1DchrAhSAIWCclAfD9rJc3PZFC\/l
w35Vj1Bg81zvUZRko39+W8FkoX7L1BWZfsJUP0AgdEAlz0BavBU3zJgSDZmJSP1CP6yZeJbNc
iXq8NK4OdUXaqoatrnw47UVd2VNGb9i9r9wT8fDZywmDwKC4L8iIeajlC8UZvsFgkCzPGZl4s
O1Fcs3RNgHGwb6V\/augh0vzDmykMYTqFte+0047rfxasnczDx8+nOyH8pzlM9YMUKfTSYypP
uiLPBfAvV8cvKQF+LxP\/avMkcew5h4cXDtMAkGThz4TJh4hHPiKhjrUJb6ssJq6Ukmz1ZVKD
tP+7imjx6B5SHkiAd41DYNFSIyN16Dgsc9Uf+DF5d9yyy3JO6K8F\/GASWY\/+1XaxIAwJAAu
LfgJ63VIA4zH2tra5peUpeXggIhRZBx5V7ysSdoXvHiXIbMIGewDBw4EyWbIC1bPFVdcccLXd
Bm9tbW10mBvFsgQsiUDEHKwoS0eYJmvL1mxmaOtrvxVMobLrCszH\/RjDPeU0TvWp8YBg\/KV
r3wlDXvgeVJ3v\/vdy9PFxgyHEDJ6DAcDEsYkQh7bkCITJTHo55xzTrLEVYfCjIlwHmC5b+k9
6m\/MjqqT50mm0fdxITr0o3jtRnqrK5NLfVpdmbymyUqc1EaPV2IZyTANE5vlpA+PWjLm+aPo
g8YDEl+J5cHxFnmVkT9JqI3bLRfN1pa49gRHLW0nqXMUrT1G+5v+hCRjO4xO37VXu4flr2qa\
/rS60nz0dklXGjXwpDZ6vAn7IzwXgxQSc4nWaat9Mp\/KtqFt1pKPDj0ceBjW1tY2P\/qpjH1
AeSD2\/yzhxAEe9hqDp7RR4I8LOazIT0SVt5cnVI5nx7BqJwMobRgEL4ZpWP64NMtNhycOLux
rjqLFe21t9HJ8VLllT291pfkILbuunNRGzzBaYjqo8BHE2FNyidaSl6fHo7nqqqsSD0e+QwjL
VWWBh8FSj8GRb4\/Naa1PY0c+D8z+iXygLt7SOOOhLNAGbdEmZYE2vPrVr968XKwNPDz7VAyg
csMAL8ay7rkOo62nOXSx+S\/Uhhxy4423tmhTnceqx1tdaTaCdGSZdWXPGD0GZLu9IErr3qCH
Px8+6fkeU52G4XMggkb4gAc8IC9eGh\/pkX\/BBRcky03lEDIA2iY\/QHvzvIgPS9NebYqy6g
re6AOaGBtG2aVQs7Fy0ba8fukg6iUfEPXXwyiLJ968Y+WXFbTXeOj7qDbqL5mTQU4jPe9\/nc

a4GB80wlZXPpnIIoDsyXM3dWXPGD2CnBD2DPkkCuShdOHUbDxrAeCJtzpmzbvlNxsJtLqSUmv
0ZqNLu8bF9RHLbUvqpsbG3y649tprj3uLZKcdsL+IJ9475dWWn48EWl2p5NoavUoOE\/1a4tS
XNRMxmCGxtlg6CJuytax75zvfWS7Lm5bZjo7BxRPv7WjlewDzfcEcl4dmL4BxaXXl+JGcVFeO
L73zWGv0di7DPcth1h2ztHLxmIHDm7EeBvLQOOHmQYq30EpgVhJojd6sJNnyGSsBBu\/SSy9N
TrIZOh7QqALy0DgR96ciR9G16a0EppFAa\/SmkVpbZmIJWPZa5gmbFkarTFP6lq6VQBMJLNTo
2auxbBkG8po0uKXZGxLg+VnquuMXOL2QJr4SvWwbuZISmLvRo8AUmUKTkGXLMJCHpt3HIYm9D
95Qca\/QvS1XXfTYRxi8ISNPvIVWAvOQwFyNHoPX7uPMY9hWmye98IaKV+a8tuciNwNoOetdZ
XloVruXbeuXVQJzNXqU2J6MsKkA0CrTlL6lW20J+BCD94YZu9XuSdv6EyWwnClzNXrL2eW2Vb
stAd8z9F1DX6vxOSyf9\/J6nHbt5Xd39a+F3ZfAQoyer5bYrzvrrLOS0Nd27d3BgfxRonDAgU
aY01gW+RKKvPpmuLQc0KFX3r2vvO6cDo4XuhwstWJfEk3eFnzxlw7G9SXnOStcfeod1m511Nu
ONoe8L+gDol9kRWaRPmmoLB55Pd5nffjDH55e9rKXJR8V\/drXvpZ8hKHb7SYfVqAfk9aDv34
JJy0b9MNkGWnGmEyCNuT6Az\/wA6VO1+Uf+fpOBgCujXVQR\/DV\/np+xOmgd4ZzPuhB0NTDv
F1BJ4z6djscJxd9yds\/q7bO1ejpkBn8L\/\/yL5MZ\/V\/+y39Ztttnknxlx IGGdIo+z4GwQ
e7hypW2bMiUP\/agglcsz6ZkNZdiFCV\/kMZV4tAA\/Tiaep5x9UdzQgbyh6VJN9a+Qxef2JK
mbS95yUvSKadU6re2tiY50RNLXTItE+b8o9\/a0qQaOgSa0E5K4yCnSTv+9m\/\/Nj3\/+c9P
\/rJg1GH8\/vzP\/zyiJ4T0Xj9PyFiRBF83oluzbO4ps2SW8yJos7cH48lPfnL5WfHv\/M7vL
El8dik+gWTW94d63v\/+9ye0JcEOfnxeiXIyqMAf9PEHdG688cbkk+rB2vuqDC6aHJwmBk09x
Me37fIHEw+f0anTLir+67\/+6+VXLKLdJg+KXq+\/Lhd9jvdkLSnr9KPiMa75gzcsTXnK6oJx
HFJIQ+shh\/tklnYAY6aNZOlBlj9PGCWncXX6A1I8uHE02+XV9Y4MlInDmzBgPsQq3RiRD\/B
JMQ6CCUQ8yn7kIx9BmtACEXno4JOML\/rdgLpc9E\/76Zmvgs+yTacYfG5k7r6bdaRRUJUFjT
QgXzqgBNxu8MIXvrB09X2+PB48jb7kkktS8FIGvPnNby5p8XvTm95U\/i2KeczwltT3vve9y9
nRBy7VPS3wWBg9DyZjh08olAESD\/DA58sQ\/STHyBeKSwdwELh8chb\/7d\/+7fJPTsKBdPk
ALo18lQ+DwbCo\/0\/\/9E\/TX\/\/1X5de1Ac+8AFFSkB7+eWXl7hDBBPOT\/3UT6VOp5PI7
N\/9u39X5v3d3\/1duQQ1vg6YYlwZKRPXv\/k3\/6bMRxxp2iK+vr5eyl1btFGdIS\/f9cOrc
6w+Sn+b29xGsfTud7870Sv02qLsD\/\/wD5f6cuWVV5ayOPvss8t2yqNv0W+hchgJo7x60KF\
/6lOfmtCh0TY04HOf+5ykEkKu8hmgH\/qhH0rGXbwk2OaHfvNcfUGbXAH8gx\/8YHJqrX9YkC
Fjb1\/Tu8vBn2MgPwd7oOLKC6MsvuLAWArJOegYSuMrPYBsyALoa6SjYwukB4hLDxr0kSes5+
sbfZEH6vnBZ7sw+hL9Rk+38AzQD+kB49oWz+QplmoKhPeic4TEq\/FVYExCQdABA1PvCIV4y1
veIjv9vb\/398pw1A\/j9prXvGYzmyLkA7eZkSHaEB0VGnAPWUYyFOXd8fKiP0GkLB54Behr5
I8Kfc1EnoEIWVHa\/LtpBt1Xhhl8tAG+uEzw4gZLn+AA\/q53vQt6AvzyL\/9y0t7IIH+DH\/
Em4draWnr2s59dGg79Vd\/a2lrS9p\/92Z8tDVe0DT+eQl4n42dykpfDf\/2v\/zWPbuLa945
3vGMzDlGndGPhjxlJs8RdW6uWt75DKM0fZGKgyVccSNMGkw5D\/s1vfjMxRuBLX\/oSkk1Qz6
Mf\/ehNwyZDPUFX10887nCHO6T9+\/cjTf7gERmXkY2ftbW1dP\/737+UVT6GG1mb\/xlwcg2
g32trVb82iYYgJk\/PzmmnnZZO24Dv\/d7vLansc0L0JXh+7GMfS2TnuZQXB0JHjhwRLUEeGs
9w6JM\/eWBFVRJs\/EjHdwMt\/+srfRQpiuI4XZNGD3xIF+4ZQQ8PkK+cON1\/3OMeV04Q4kA
++eTjKX0coA3bFMaP7uCTl9OPaPt2bYtypzAIIgRvAMJI8I5uvfXWZHki30PA5dQBD4oQyAvg
WqMxU6OXbklrdo7llzRKRvBolZF29OhRwY5BPw4cOLD5cFtiMz76YzbfaQWhVAaEEpKfk0gPT
fB27caGs\/4BctUm7eBt5gOq\/2jIUn7wyENlP\/rRj5b7onB5jK4wB++s4ieN5+kjlj\/4gz
8oOhLI61\/\/63+d9EMfPBwPfvCD07\/\/9\/\/+uDKMzF\/8xV8kG\/cytIMHry+MZqRFX9\
/whjdIKg2JLQb98\/AxXvamyAGBNEBfPCweWOmMIs9rba0yHNJ5Qre97W1TTJCPfexj09raWr
l1YlyUe8pTnlLG9YdRiA+aytc\/7faA0gf0d7nLXZJxBKeffrqkdKc73amUtXEJI+RZeMELXl
B6rvYpS8IGPzy23\/md30kAvra2Vh7WaAdjFg+xLQD9cTm7AduhJAxA6Dt5qxNheHxwQM\/k6
1\/oS+hTiaoKtAAAEABJREFUbJfIA\/V85YHxkh+gnHQGkU7Rv8iDqw+gGQbycicELo2tUZ6e
xZ9hiLoj\/+1vf3u5Mgi+kR\/1R9si\/xQC8HcPKJWOWz9Ls\/ww28BVGkbLAx0Dw4UORso7t
Ij4uJDiUjY0YcXh44DwoxPCeLjGlYk87bc080BHGqXDA6+AunCCNg+\/\/\/u\/P3lgeMaMAP
mQVc476A0UjzgGMNJ5usrHgErP5SqeQ\/BXBzzPa4qbaF7xileU+3\/6S57SlOfh6ce5556bv
uM7vkNSshQkozKy8fOtb31r4zclX0aG8M5\/4zd+A5oYpxI59hP9E+WFeBDx4n0weDw2eYAHF
XWKmyR5ZX\/2Z39WLsulMTxkyVCGztGfM844Q3Z65jOfmcK43e1ud0sMiAwGNgzU5z\/\/+SR
O7tonHzCG5AoPIONIo9eRrg\/Agx3OQOTVHzRtIl8G2tIWwIM+D41FXffoGBp5xouuftd3fZe
kxsDom3Di2Y2C5EPfxEc9f7wnRpkhRRcQBpSxlg94YJHPhsDJSB6AS4s8eBPw3HIetNWYGTvl
om5jwcDaGqAb27VNWXCKQTXDcq19BBJIa2rAMAHKUyB4Hf7Fv\/gXpecVe0iUT4MJ5Bd\/8Rf
r5DuKMyRmAIoSUFeonVTAW+CZELaZH6+64nDx7WlYClsqUVj9RTsN1PlPw0MZXrwQUMC1tbXE
+2BQPKDSAeWiZHDpDCJ8FrBv377Ea7rrXe9asiPLEtn4+aM\/+qNkCapOY\/fxj3+8XMI+8IE
PLJd1Zvr\/83\/+T2mYTZwbRXb8n1dXZzJK3gzhZZddVrbF2NOBetnt4vqmz7mO8tC3K6du5Y
yFBxy9MTJWsSxnIPH1DHiGeUC+XYc2h1H9Q2OJ6LlEYwzwlB7AgEpn5CONETK506NImzT0fHh
OAFz5ONyBN4GmbTvFjKIS1t+SAEizFGTECI+lDmvugSZMjQjLCq9DlP2RH\/mRcvlAUM997nN
LMqE4iGUQXix6SbDkP9oaTaRgPLmIC8nLA0Gm+hTKKQ+YgS2J0aCVlstVfFqgrKPKDgaDMktd
FMoDyPtg+DxUH\/rQhxJPzMz5+7\/\/++Xejm0H7Y9DBg8RWktgS0JjGIagZL7xE\/3bQDeXt

+hA7HWFATb+5ODBeerGAQMvkEHkUWonHjw9OumB5i1ahgrDw8STR4eWR6dvcBNUeIChc+E50G
s0k4L++0NS25WzzNZusrXMBHBy463Si+141PNjbJWXx0CQHb7iOYxzQnK6HOfx82BznTY55jS
Bh4ExJujZDWMabeSlGe8cmhp3BpVN0rdLL720vNWR65T8nK9tHOOyXdsi\/xSzByXOH+Rw7zG
y9EVMKc0A0RghkDcOCIVRCKM5jnZV8sKgay\/jdfvb3x66CTHwDFousyAgbzIWt3wImmkfRHz
qoO44vWWw1tbW0n\/\/7\/+99LiNG4VaW6v2yyxrjTXjxgvnTbhHh+ff\/M3fCBLPygTmwbDt
EeNqcvzDP\/zDkibShOHBx\/JWHwFvzSFDWWDIT2fjNPef\/tN\/elyOukNeMv7xP\/7HgnTN
NdeUniAPI7wfp6bijKT+hDG0ylA\/T6YsfOzHMp2c9OtY0raBB9qDvi3hjAmsvuiIsdUXz6Qq
fvRHf1RQAjmYtIbpZUkw5odeM5Z0g46oQ11RhFx5dNID0KEnb7oQslEuaISTyNjzERMpXXKII
s3VNm3RbzwDtEm\/hZEmrLdNWXBKGDsdJlBAuDKBpWHdxdWx+h4Z2hwI4J73vOdmkoeA4kkQa
hSQLs3sFTO7+DKDvhlkbfQwGhB4gFmQjCIONzuJ66fQrJfLFf6TP\/mTsnYEJph4IBkvy8VhS
7ioRD8swbWPkkjnbQkBg0M37LPZE+S9rq1VxlK+vv38z\/98uc8pDngHZBAHHNIA\/upxsKWc
tByk8cR+\/Md\/vFxC2t+L\/JhI9I3XRl7atLa2luzvBZ1QnkMPoXgA\/vRZ3P4YTxdOTh4ae
BPgpVneNqGdJY2JiQHwjAZffcyfM\/3Qn3Bmgq5JWO8XWfNQhbxnk4jnnhxzfvLpkOcADx+Kl
RY02qvd2h9p24Vow6OOE3M6RX\/ysqFTdHS7tkW5UzASUQk3EcClBXhAc3cyFEe+TlJUAJcGC
MB1hijH9aWshGEZFelwaYSWl8cDRN1C8QD8dRIffVBWGwA86Oqhvumjsnju84fF8cMXwJVTXt
3RLm0QD9kIxQE8z48Zk5dHgdC4eOpEOK9fOXnKRrr6pAml1Wm0TzvRACeCeVxaDowDPtEn42Q
ZHDSuh\/CG8GDs\/8t\/+S8JTeR7sybKRlq0rdfrlcYr+khmaNWnnAd2ba0yoJSZYTNbq8cY2
dNTF35kgL926KM0ccDTVAe+IB4SNMqjIYfwPqw6\/K3iyNMO3oN8uLrwAeSjvPaKB6CR7sBFK
B55Qm3UVv2gcwCeywBdHbQZv6hXPKcJPmiAfBC4CQd9ODPwgIsvvrg8yEIfadqtLD2SFnFp2u
9+pBDoExq08gPyPPnopEW+fmu3vDpIlz9MLtqJh3x0yubtk1cvt13b8DjFzyKAF2cv0AxJKFE
nXNosbrsHz2UPPfg8RO30oPF4eWhcebNi7mmjWVWgqLZHeAnGP++HZSYj\/3u\/93vl\/Td5D
L\/QlRJhU3DSaI\/PHpcHgYfgnmTUychpA\/kyrrmuWdZ6cJxyAri0pnUvE53JlEx5P2CZ2rZ
MbVmY0VumTi9DW8xiuXeiTQzepMsA5ZYZHCTYY2Lco52MESNkj8ZhQ6SbDKTJQxPp24VkyVgp
j9akYelqqSduS8Gy3ATLGNjvYiDluaLlwjPvkpF2VUeavFUDRh2QYchi1fqwiPYuzOhROPf78
llWBym3NHlopM0CzP48KACfBc9Z86i74rkbP+u6dosfQ6JfDNNutIH3wwu0dM7rj2WgsL7\/J
S2nXRU8ln7CVWnzItsZujja6M2hNRTfUtZsyxgBuDR5s6rS8sTMb7kC4NJy\/m78mxVbuL68m
jJODmSVy26nuMnNJOck12lx8JvVBOjENry34B1GjwfEyEW6MPLquDjQ\/3HyafOO1yHyIrdlh
YUaPUIwC9l3yUGavFmB5UkoPeteX7IYFO+h2jRv4YnlC\/zj5EBWZDar8cHHJGcZ5rt6rjw4S
DABMoby0EwDvHp3zVx5YOCm4ZGX0W\/9HyefNu94HSIvcsvluEz4wo3eIjpveWI2z5VeWtRtQ
Jwau4LBC9wp+Nus9q7cEdspryblF1mf92vJisxCfrMKTXb55AffqcFzcjtsXzTGP5a7eR8iT1
qOi+u3\/s9KV5qMbxOaRepAk\/YEzTz1xXjMAuZq9CxXzILCpo1Fq0xT+lF02y1ZlPO1De9+T
gf3T\/e\/\/\/1LcK8s7qrd\/1ha0\/Aud\/nRdOTIgfTxj58I73nPgTQM\/uRPHpu+\/OVfS
tddNzx\/WJlp0z72scdutO9c4poZMDzG2KnqrJjy8NwO4OXx7oOvic8EGPEIQz8ijHThsDSHH
zzRZQHvM3vrw8Xksk1HjqQD73lPKvEDB6YP8QBT8nDRnQyXGeZq9OzduH\/n3V77d+OUnNKic
cXB0mQRQnOq6FWmnYJXn774xS8mp4VNeH34w19Mz3verel2t9uf7nnPfenHfiylpz71RCiKlI
oipaJIqShSKoqUiiKll7xkX7ryytPLsChSKoqUiiKlokipKFIqipSKIqWiSKkoUiqKlIoipaJ
IqShSKoqUiiKlokipKFIqipSKIqWiSKkoUiqKlIoipaJI6Y1vPDPdeuuBmQ6HS67eGZ0VU3rF
w3P3js7V+Z5\/\/vmJQTSh2n9TtytC6Jz0uudnzxfApckLOPPo0XTmk5+cUr8fScsVDgapVKK
iSGl9ffq29fspFUVKRZHSr\/3a9HyWvORcjZ6+U8K4qHjdddeVr0ExbnVAa3njpC+fqaVPA\/
kyJcdzXm7y+1LHTsFtcO+L+sLHdrx+8zfPSOed9z2l0crbcjLhdMLenYMMXt9O+06vvIXA+8n
1ijHE2zLalRVekL1DB2faII+u8TqVBXBp8gJevjGhnfmJT1SGxewUGcsYHjo0m1YVRUr9\/mx
4LRmXuRu9vL\/1KxqMXMDFF1+ck+4IH7Y8GZbmix++ULFTuN3tble+m7odn8sv35+AznU6KfV
6Kb3pTSm9730p3Xzz8fDtb6c0Cm699Uj65Cdv3vDAjoykGVV20vQPfvD6dPrpV2rypDCSnsfF
8+J18bhyQwX3hRo0IxnUMkbpFWMXpDlNni6f7n3yk5\/ckOknE1xaDgc2lo6b8fX1lM46K6XB
YDNpqZD19ZQGg9k0aY96ews1erMZie25WJ5YpliuALi07UvOj4L+FEXFv9OpjB2D1+ul1O2m1
Omk1Omk1Omk1OlUdHv1l5cV3n8YmzyUh2Zp+z8YpNLw9fvL2cR+fzbt6vdTWl+fDa8l4rInjZ
7liWWK5QqAS9stua+vp1QUVe2dTuXZdbtVvP1dIQkURUpFsdVgm7Fms62U5cBmtcTVG8v5wQC
2Z2BPGj2jY5kS3gNc2m7AYJDKQwp1dzqVwet0xE5esHS1hLWUHQfez106Kb3whdUgRsOKIpWH
CBGfIJwb6fr6zlkXxRYPhm8rtvLYnjV6yzAyg8GWwdMey9lOB3Zyg6Wrgwxf17GvF5MTXJovn
Xg31utjS2n4ut1qA7bbrQay31++5e76etW2aX8f9rCUut2qdL+fUr9f4Xvgd2WNnr06HyYMT6
H+cMQVGPnw3Rirfj+lfr+quddLqdut8JP9l6fnICM\/RSUTxlCaPNdavLHh4wPylg46ncrjK4
qqaYNBKj2+ZVnuzmKJ64St6l3Vt8BXPFy40WOAGCLgSoG4PbdJri54aHw6yDfOeAk8BA8HXsa
DQXRD3Hu3AC5N3qJgMNjy8jqd6uBiUXW39SxQApa7XPiosihSafwivlvhYDCbmnfT8M2mBydw
WajR44353tdOv6HGI3DCF1cPxN3Divt42717e4IUZpwwGLQGb5xIjZflra\/rmMCCFi5NHhrj
6U2IyF\/asNerlrudTtXEfr9a7lax3fnt91MaDHZed7ebUrdb8en3U+r3K3yFfxdm9Cg0b8yS
ZRbfUBsncw+LV4+8ghR00gKfd9jvp9TvV7V0uyl1uxXe\/m5JwOGSpawLw7x+AJcmj9fuLYtF
vZ2z1bIpsU6nWu52uxWDwaAyfP1+FV\/Ub7e7VVO\/v4XvBNtj3t7CjN5OZL5dWUtX71x63Sh
o88vIOR758woHg9bLaypbnrrtiRykKc\/wzertHPwWAp1OZfiKoqpuMEjlUrffr+KL+L3wwq1

aZrGvF9z2kOFbmNGzXLFs8erRLL+hxoO0v\/fIRz4yxQMT47Rd6C91zQq++c1vpltvvXXzeoq
6X\/vaI2lW\/Ot8or56+qzjR48e1ZUWJpFAfZ9v0ff5ut2qtYNBFc7it9tNqdutOPX7KfX7Fb
6CvwszemRj9ra8nfQbavYCLX9AfujB4GBebIwAABAASURBVPmYgf08rxmpIyBfzuZ45Atn\/c
GB17zmlk1deOxjv5Huda\/PpSYfIJiUZtIPHEzKP6e3\/0pWLUwogV4vpdw7KoqtJcCErCYm7
3arIv1+SoNBhc\/iV386nYoTQz4YVPiK\/S7U6JENbyxfzsAZQ3mjgEFDB5zE2qsLg8d7lJ+X
HbacHZY2yw8OfPObZ6RXveouZTM6nZSuumpf2u7jA9PmT\/KBg2nriHJnn3122ad5\/Ni3M5G
BaU\/y59GumfHsdo8\/4FhfT+Vyd2YVjGB097tvZfT7W\/gssPykekUvLS\/c6M1C7q63XHrp
pYmHN8xges\/W+7b2+gBcWr3uWX5w4HWvu1v6\/Of3l1VY3Wz38YGd5Df9wMHxdexP08Tvnj9
AZe9m88N7n8VJ\/mxaM0cunU7l8XW7VSX9fpq74ev1qrr8znJfD79uN6WigKXU76dV\/ATVXI
0eb6zJ60ZmenToK2mO\/7355pvTjTfemPyFLWUDYunrPVu4924BXNp4rtPnXn\/9\/nTNNXcs
GXS7KfV6Jdr+jJCAcV7USf6IJiw2udM50fD5Uss8W9HtVtwHgyqc5a9ZvdutOBZFSv1+ha\/I
71yNnsML9+ksS4HXi7xm5DKxOIBLc1UBfRO5MWBO9pTPIZa+ePAAIw8ubR4wGKTNw4tOp72EP
A8Z7xme9sS63ao7g0F1paWKzf6326149vspDQYVPsvfFV7mztXo5TI2u3u9qG7cGDppLqWiyc
usAn7wYEqDQdXSXi+lTqfC29\/REjDm9mJnfZI\/usZGOYshYvh6vaquwSCVn6iqYrP9zbcl+
v3Z8sat09ma4QeDNPcluzpnBAszejNq71KxGQxSKoqqSWeeeTQ997lHqkj7u60EeN\/TnORvy
3gKgvxABT4Fi8mK8JJ6varMYDAfw9frVfz9znpfD0\/Q66XU7cJS6vdT6vcrfMl\/F2b0Ynav
e3S8O2lmfjTTyMvJX31PkPLGXh98Gr7jygwGW8taBu\/lL\/\/iOPI2b4gEpjnJH8JmR0kOum
yLeEcbwKXtiGmTwos0fOvrTVo0HQ3PtdOpyrrGMi8DW9Uwk9+FGT2tNbtbynrdKAwSXJo8NJN
CGM28HKWlvJQYwKXlNDvF+\/2U+v2KywUXHE0HDrReXiWN1frd1fe05234fB4qhmMwCGzqcGR
B\/TiWeeYb33gMW95goUaPGIbN7tLkTQNOcE899dTjis5bkQeDLS+v00mpKI6rvo0MkYDJiTc
ek924EB36IWxmnuTi+m6+p13+kZRer+rXYJBmujfW61V8\/fb7fucD3W5KRVHy9geULvnyl0t
8WX8WbvRmKQje2w033JAuuuii49jOW5HzO5nZJHdcG9rI8RKwdXFoY+kTJ+qzOsk\/vpbpYvn
F9RzH7cDHPy4o4ei3vjWf1wpf+9p09KEPLetI\/X5p+Jq8ThivIuavCsLzsqnTKfkefe97t22
7siXxxk\/OoxH+3Odu9uGSr3wlMX4bbJby\/8KMnpnbDD5qhpeHZhIpHTx4MHkNzRsa9XK58u
Z4nW7S+Pp6Sv1+VarXS6nbrfD2t7kEjPNKnOQPBlsu\/Ub3vv6Nb8zltUKv\/X36zW9OR8499
kfV+\/106\/OeN7au+quIG80r\/395w+DgF\/CNBz6wTN\/31remL374w2N5+tvNJfHGz5c3v
LXg0Tj8\/\/6\/dHjfvnTlaaelw\/e4xwaX5fw\/N6NX7259po8Z3z29e9\/73ulVr3pVQlMv
Nyr+3o2ZS95OlsaNZrAjWx8N+MQnjm5eQHd4kX9QIGbdSXlOS7+o+vLZn7z3IlgZRL9yfPMu0
rHMU+94x7m9Wui1v7Rh7GIWPf3KK9MZN900sr78VcS73vWuKf6dvmFw8ArYf\/HFkTWWH\/rT
Tz99kxZPaZPAXR\/0oPQHV12Vrsz4bDJcImRhRm9Unxm6Sy65JL30pS9NXi8bRueVpfAQvV3x
hS98IV199dXpwgsvHEZepuXKm+Nl5rGfST848LznHdl8Di655Mubs2Z91m08M35uug8SLLK+e
XxwwJg\/4QlPSE7teX3HhiPBpclDE+nzDIetAoalacO+29wmTfMq3yRlyj0+lW3A\/p\/4ibT
\/+uuH1ll\/FXGDvPxff7Vy3yMekVKnU+X9wR8M5ZW3ryTc+KnzyWnG4fN6bXGjSTP7v+tGL3
rymc98Jvm7CBHPQx8UCM\/QSayHg4foFTPG8GlPe1r69Kc\/nR75yEcm+3zDlHZY2iQfHLjpp
jOOe9XsWc86dXMWzmfdSWbGaWkXWd\/Zc\/rggNN6p\/ZO740hgEuTl4\/\/PHHvZHs3m94Au
LR51jmWd6eTkmsgQZRvIEfapGG3W5VYX09pMKjwk\/h3KYzeddddl3wW\/Pa3v32joai\/hvb
6178+eZXNh0TlUVrKS4kBXFqdedPZzIcELr+8+phAp5PKyTif7eqzbp43Et+\/f9tZd1TZRdY
3z5nb1kRMZhFKq4\/TPOP0xerBBArg0uZZ57a8u92UiqIiGwzSjk90L7yw4uXXElp4EsPCjB7
vzGGFGb0OjJXPgg87kJhmbCgt5aXEAC5tGl7K0BMA7\/VS6nRgLewVCfAsw+jCl6Jf+Uv9lG9
9ffpmdbspdTpV+Vl4jhWnlf1dmNGzR3Mou7IQSib08YCdGCXeAd7qiJGgvHgDeKRPGg4GWwd4
nU5KdHFSHi19KvfrTD4mv6byQKtMU\/o9R2eZ2+lU3WKsGL8qNvlvrrg74TN5zUtXYmFGjwLz
9OLUNZeENHlo8vRlwOlatKO9kxeSmDw0IV1xxRXlFSOevjEfxcVrg9\/3fd9X7tFaAaST+V+u
dL\/2a9NLotfbKnvw4BZ+EmILM3qrKFsTItD2Xi+lbhfWwrQSYPgOHfP27eMyfsMAfx76TlcA
+Kw8dLspFUXVDco4C8O3vp7SYFDxPAl\/5270YtZ2MueE1UlrXdGl+Qqyh2KSMQje+FkG5Vde
8jz4JHzRDgbHL2vzCVd+CzuTQH4iz8DlsJPtiJ21aklLW5p2u1XjiiKlwaDCJ\/3NDzQOnrze
3tyNHgWm0K6YOGF10ipeBw\/BJGNoeeTvovqgAF7enyyKomThxNbVFnkALq3MbPhDJwaDipjO
VVj720pglySQK2G15zJ5Q7rdlLrdqlxRVOFJ+Dt3oxcy5cUd2ljaOHSItJ2ElkfPeMYzUhyAM
JoAz51+cGAwSKkocEqp00mp16vw9reVwK5JoNtNqderqu\/3U+r3K3zS39zbW1+ftPSeoF+Y0
ZultBx4+BsLwy4cq8cbGDy\/\/AqMNHlNICbSTqe6k9ekTEvTSmDuEsj2WPZvTPhT1dfrpdTp
VEV3sj9YcVjJ37kaPcbJqaylaOD234YBOjRNpXinO90peR3tfve7X8KvvqeXG8Qcz\/kPe+\/
1jW88mvr9iuohDzlafidvGF09bVHvwka9i6rvZHj3thrtFfkNwzcYpDtec810jY6l8gaPTWWf
jtNSlxrVuLkavXxJG7j9t2Fg6YtmVEPr6V\/72tfStddemyxlP\/rRj5bZRVGUYdOfr9S+SPH
hD3\/xuA8KvOhFn958v3bc+7SLfBdWOxZZ3zzevW06Pi3dEAn0eil1u2XGXZ\/97DKc+KfX2y
pyEnp7czV6W5LdGVb\/4EC8o+tvLFjCArglb3iL+XI2x\/OW1N+9vfbau6bDh\/eVJCbDpu\/
GLvJdWG1aZH3zeveWkJ2q89LBe9\/73iRe99jRtVCTQLYvt\/\/yy2uZDaNFURH2+yn1+xV+k
vwu1Og5QY3lKEXPYdzy1gFFeIdOYhkr7+rWx0ia93eHLWeHpeXv3tbfr73oon2N341d5Luw3s
ddZH3zevfWRHZS\/LHvNId\/vV5K3W7FuCiqcNJfs3qUOcm8vYUZPXfofD7Kl1DCgOXhJMtbn
t15552X\/AlBfAFcmjwfF\/CRAUYWwKXFGA8L4\/BCXmybwFuYvQR447xy3vkd7nCHzQqMnTR

5aDYzWuRECWTeXjo45Z27oqj49vsp9fsp2eOrUlLqdALbc+HCjJ4lqc9HnX\/++TMRovt\/jN
x973vfBJzWSsPcNRbLJB8bAHBp8oZBv59Sv1\/l9HopdbsV3v62ElhWCRx5\/OO3vrRcFNM1k
7fX6VRl81m\/Stmzvwszepadlp+zlCQjF96iJXDOO8+D53k5bnIL777TaT8okMtmXrgDKx8K5
Z3fcsstm9Xw7hb9EdHNylcQ+frP\/dzOW83w4eJBmNZjVH6FYGFGL5YulJpyL4uM+v2U+v2qN
b1eSp1Ohbe\/NQnMOGoispR9+MMfXn4A1quIXlVkDOXNuLo9ye4bj31sOnrmmTvrW6+XUqdT8
ej3q3CP\/y7M6DF0DJ73byl3fogBH3eQMa8x+NSn0uYVlU6n9fLmJedRfB\/xiEek8NQjnNbg
ORihR8Bhmb3cvN4830lxnve6172uvOupLDzPW3b86AUX7LyJ4e3tnNNKcFiY0bOkOXTo0AlKH
souD80kUssVGZ6XpbyUGMDzvMD7\/ZR49eIn2bjr8q6CSdBExwAFbqykiU\/SOOP72c9+Nrmv
SZ+8nvisZz2r\/IYfPupwOOL9b+9+m3yjDsbRjQDvaAO4NOVWAori+GZ2OsfHm8R6vZS63SaU
e4JmYUZv1tLKFZ0yU2pp6qG0lJcSA7g0eQFHj56ZXve6A2W000mp1yvR9mdBEnj5y1+eHvSgB
yXenj\/YrlrjaHkrT7wp8A6NsS0UZZzUf\/vb3y7f2BH3nra6TKpWGfaWGUh5LrefubFEPOus
s8r3uM8555zywru8yWAXqbvdnVd+Es36CzN6ZlazuNl8HNQ9tlGj6cKxE1uKTpkptTT0TRT5y
1++BGkJ7RWVUgwL+6ELJikn+a4bua9n\/Iyj+5Ty0MyiQfjzAvHN+YWuCEOPIl9aiXc6ZbD0P
\/n1lWkb2+2m1O1OW3qlyi3M6FFos7jPS5nRLUMAXNpznvOccnlCQZsYPkqMllJ7QDwoHiLSp
7QjFXmDgJf3jW88dgNLqdtNqdst0fZnFyQQV5mM56yqP3jwYArvLXgGf5Mk3Yh0YeTVcRv8h1
\/84nT9\/v0pdTopPexhSJYPer2UiiKlXi+lbnf69vk8fVGk1Oul1OlMz2fJSy7M6DFMb3\/7
25M\/8ccAhlzg0uR5AJzoMWCRPyq0pEHrjp4Lz\/5YuKVS0I9U5A2CF7ygOvHat+9wOv\/86x
PDuxP4yEc+kt785jcnS+md8GladpH1uVu5IbKZ\/o\/rSyYny0zvQFuSqiRfiopPCrY4ZvmHp
g5vnC4\/8Ywz0pWXXloav6ZjNG+6E3Tgx388Xf\/0p+9Yl6\/fIZ8x+jLpUM6NfmFGb9Y94A2
65+WhsZz1tgeFb1LPW95yOHW7Ke3f\/6H0m7\/5xOTy8k7gKU95SuJd+HsOO+HTtOwi63v2s5
+dzj333NJzaiLbJjS8LbLyEVhXVUxaLo8bU957URQj2dibdTobWyQOKYLY+ON51VVXlftzkS5
kYIVWBuqAB0T9cad3AAANaklEQVSeeI6L8xj1\/8orr9yxnjQd3yZ0i9SBJu0Jmnnoi3GYJSz
M6PHoLG\/zkzMd4QFKk4eG0tlolhfgYQglJ9zBYJB4gzw9DxCA8xbxUw4fIchxcYrM8P3u735
3sgHewtvGyuEVr3jFTI2eMWDk\/A0MWxxxsVxoLIwnmmGQl1M2vHsGz\/jz8tBEWbzqy1l5sR
KIUFpAnkZX9F+7WhivJyEf8iK3kOeyhQszejpuSWope+DAgc17UXBp8iiumZoXgD7Aw0DBAcF
aHkXesDBX2sivpxkUdbdwIG0nA7IKOS5jyNujN7Y4TJz1Np5\/\/vmJQTQhWnZagtkWQWdZ7d
1sHiSAS5MXoP\/byajN39Ij8grZLWO4UKNHAGZmxisHafIYPrP\/fe5zH9GRQLGd9lneWq4Au
DR5lJbyUmIAlzaSYZuxcAkYl3yZGp680Ck\/A9W0UfYBv\/71ryfvWSsfwBjiQb\/oBsNkOW2
SpSfyeIVWD8oCuDR5LexNCSzc6M1KjLHvY8YGljA8QvwpLeWlxAAuTV4TqD+Qltd5OR5pPFjw
PM\/D6qGVLxTP89HLA\/A8b9E4o1BvozZpG4DnbdIX9PKE4nk+enkAnufluEnKHqxLxPZkGSO
XhnlhTvJzo5SXG4Ub93wSDZyxizI5TZ4u32QbZeDSgHHXF0BW0uYBq6JvZFAfd+NMPgCey4d+
oJcnFM\/z0csD8DxvnvhCjV59cHU2YJhQxnXcXo2lbigrpc7pKW\/kwfO8cbiBcZv\/la98Zf
n2iAfR\/mEMij6o10ktgEsLni7W8irULRSPPHTolQNwaZG\/yFA\/7aXmdWqLNmkbgEsLGn3R
p532zSm9JaYtB+NowrLvyvti8HjtDGPUuxuhB9y4G38GmazIbNZtwXMV9E07ySDvP92gI3QFw
KUFzaz0JfjNKlyY0aPEs5zdZyWAOh8P3qFDh8o3BeSJe9A9lOJOiu1ZDLvBTzE8KPaQ0F544Y
Xppptu2nwdalxZ9JPATmm9BXHqqacex2Zc++bZN8Yv5KtBDCLDCN8tsGQ27safJ+pwjVc66\/
bgvwr6tkz6stMxWJjRo8SUmYIv6+zeRJgeTp6JPgS9NLg\/VOT1J59zFwdf\/epXN1+HQjeqL
NpFgdn4hhtuSBdddNFxVY5r3yz75kFnUBgWDaATJguGVRsYmO0Oq5SbF5igXWvRrrwObcvji8
DVOUpnZjkm4\/qy2\/oyrm3T5C3M6NUbR6EMaKQziAxjxJclNOCuQYT3pl3aLgQ5Ln7nO985h
dETiksPyOlzPPIXER48eDA97nGPS7nhjnrzNuW4fH3RJ7hQHB6Q0+d45OehPVmGxbaBPTZGkE
flFNbp\/bC25eUXgUcftIXhWUSdy6hvy6Avs5T9wozess\/uw4TK87Df4vKsB3MYzaql2avS5
t3uD0NiDyj2W+3J2itscnqv\/dvCChIso74ti77McjgXZvQ0ehVmd+0EFJA3xAPxQEoLyD3UH
JefL2ctP8SlB+T0OR758wwt266++upkr3FUPXmbchy9vugTXCgOD8jpczzyVy2MPpAbr3Se7V
9GfdPvvagvCzV6qzK7hwJ6S6Ru8GLJkz8AkTZsyWcJKB190MEDhqVF3qzDm2++uXw30zUep+b
urPmoK0\/WsmpYWyJNH\/Qlb5O4dGlBBw8YlhZ5yxzS02HL2Xn1Z1n1ba\/qy0KN3jIrerTN7
HbppZeW33qLpVfkCV1ydtmZkQBwafIs4e91r3uV7+GK2wsRly6ODr1yAC5N3iLAXUXLR8tI4C
qGe3H2LOVpizZpG4BL0zZ90Bd9EheKSxdHh145AJcmbxigmdXl5GH8d5pmD3fUWxw75Z2X3z1
9y1sxHKcTy6Ivw1s4XWpr9GpyM7vdeOONyRE9byjABWcKShHgvCUAlxZsfCLLSaRyQvHIQ4de
OQCXFvm7HWqLNmkbgEuLdumLPu20b+To+hIPk\/Gtw6FDh1IY06h70aE9T1sbDld4xO4PzqNN
q6xvdIOO0BUAlxZjNSt9CX6zCudq9Cg3QXhItoNJLyfPSgB1PgYtn93igbTpbtmDngcY6XBpA
R4MD618oXjkCdHLA3BpuwUe7HobtUnbADxvm76glycUz\/PRywPwPC\/HndI7redN5enLhtva
0BdAVvNo3yrpGxnUx904kw+A5zKiH+jlCcXzfPTyADzPmyc+V6PHSPj6SXTAckoHh8EwoUS5N
txbEnAHz128vdWrtjerIoGGRm\/67pgdwsi5jBoen\/tZ03NtS66yBGIy9FqTTfxV7kvb9tWT
wNyNXi6SWC54ncffRQgD6MXunK7F954EGDdbGDHm9smcHNszi7QI0aHfe1Joe7QMElio0YsOm
+ntkfEAGUB3oFpFD+nszdB+ji0MY74doEO\/NyXR9mq3JbArRi8\/4IjPQq24ou\/2OK5M\/b
z61qNbmeHakw1dmNGrGzoHHDHjW\/buVLpel4mHqR62XuTx0rV0fPSjH53clYOTD\/kdT5VSj
FmT\/Ve88MSvzifiDB6vnndv7F3+9tbLuDJRdpahvtZ1JOJksej2zLJvs+ZFFsbV+MLJh\/zq
9cxaV+r8Zxmfu9EjIArloqoXySk7cMAxy47g5aKt75\/hn0PrRZLOFvjO2aMe9agT\/njOFsX

kmKsXeOI9rLSHgsEz2dneQPOkJz0pOcVlBMUXCa2uNJO28TSuxrdZie2p8MIT7+2pZ08xV6NH
0X0QUrOHfc6bMQwwg5hJ0LYwPwmYsX3j7zGPeczMK8ETb3XUmcfdvHq6eLzjCm9heSRgHI2nc
Z11q\/DEWx2z5r0dv7kaPTN6HFjkntcwfBHemGXa05\/+9PJP+TG24gQU3qg0EOnyQJ7POL\/
1rW8teTDqBu28884rl4poQT0NXX5JG48w8EJxPIXqB3kb6uW9vqUOUK87+GmzttSh\/vpYPX9
UPPhqWw7RFuUcPpjB1SG+ykD+ra4cTPmrhk3Hc9l1Za5Gr6mQFknnPVNLLIbXLXDK7b5YLIuF
3rmUrl2Mx6\/8yq8kn8NWxutIQF4ToAAMAeOkPBi2l+Xiti9ayIf7rhyjxuC53qEuy0D5\/ra
ET1758rGvOPvisXyARuiASJiDtniNbJo3IRg0E5P6gXpcN\/EqmeVK1OOlfHWoK9JWNWx15cN
pL+rKnjJ6w+595Z6Ih89eThgEBsV9QUbMQy1fKM7w+fu6lueMTDzY9iK55mibAONg38r+VdDD
pXm\/N9IYQnWLa99pp51WfnHZu5mHDx9O9kN5zvIZawao0+kkxlQf9EWeC+DeGQ1e0gJ8Dqr+
ZefIY1hzDw6uHSaBoMlDnwkTjxAOfHVFHeoSX1ZYOl3ZEFSrKxtCWMD\/PWX0GDQPKU8kwHu0
YbDIk7HxGhQ89pnqD7y4\/FtuuSV5R5T3Ih4wyexnv0qbGBCGBMClBT9hvQ5pgPFYW1vb\/Bq
ztBwcEDGKjCPvipc1SfuCF+8yZBahh\/DAgQNBshnygtVzxRVXnPD1ZUZvbW2tNNibBTKEbMk
AhBxsaIsHWObrS1Zs5mirK3+VjOEy68rMB\/0Ywz1l9I71qXHAoHzlK19Jwx54ntTd73738nS
xMcMhhIwew8GAhDGJkMc2pMhESQz6Oeeckyxx1aEwYyKcB1juW3q\/6lWvmuhLKDxPMo2+jwv
RoZ9H+6fl2erK5JKbVlcmr2myEie10eOVWEYyTMPEZjnpY5KWjHn+KPqg8YDEV4V5cLxFXmXk
TxJq43bLRbO1Ja49wVFL20nqHEVrj9H+pj+PydgOo9N37dXuYfmrmqY\/ra40H73Z60rzurej
PKmNHm\/CXhrPxSCFsFyiddpqn8yn1W1om7Xko0MPBx6GtbW1zQ+HKmMfUB6I\/T9LOHGAh73
G4CltFJx11lnJYUV+Iqq8vTyhcjw7hlU7GUBpwyB4MUzD8selWW46PHFwYV9zFC3ea2ujl+Oj
yi17eqsrzUdo2XXlpDZ6htES00GFjyDGnpJLtJa8PD0ezVVXXZV4OPIdQliuKgs8DJZ6DI58e
2xOa31KPfJ5YPZP5AN18ZbGGQ9lgTZoizYpC7Th1a9+9eblYm3g4dmnYgCVGwZ4MZZ1z3UYbT
3NoYvNf6E25JAbb7y1RZvqPFY93upKsxGkI8usK3vG6DEg2+0FUVr3Bj38+fBJz\/eY6jQMnw
MRNMIHPOABefHS+EiP\/AsuuCBZbiqHkAHQNvkB2pvnRXxYmvZqU5RVV\/BGH9DE2DDKLoWaj
ZWLtuX1SwdRL\/mAqL8eRlk88eYdK7+soL3GQ99HtVF\/yZwMchrpef\/rNMbF+KARtrryyUQ
WAWRPnrupK3vG6BHkFpxc2CQK5KF04dRsPGsp4Ym3OmbNu+U3Gwm0upJSa\/Rmo0u7xsX1Ect
tS+qmxsbfLrj22muPe4tkpx2wv4gn3jvl1ZafjwRaXank2hq9Sg4T\/Vri1Jc1EzGYIbG2WDo
Im7K1rHvnO99ZLsubltmOjsHFE+\/taE+mfOPS6srxI77butIavePH42SJtf1sJXDSSqA1eif
t0LcdbyVwckqgNXon57i3vW4lcNJKoDV6J+3Qtx0fJ4E2b+9KoDV6e3ds2561EmglMEQC\/z8
AAAD\/\/3jVgFoAAAAGSURBVAMAyxmN4V35rn4AAAAASUVORK5CYII=","height":100,"wi
dth":101}}
%---